

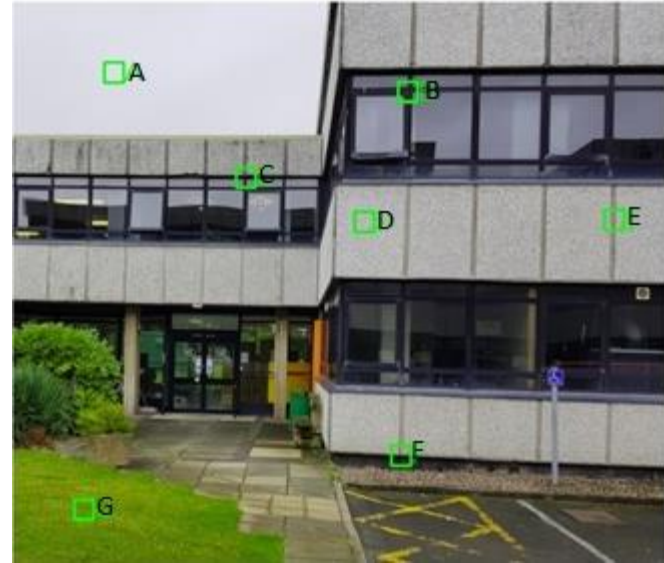
Camera Geometry

Cameras, projection, and calibration

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Quiz

- Select all that are, usually, involved in creating image panoramas:
 - Detection of keypoints (i.e. interest points)
 - Description of detected keypoints
 - Matching of keypoints using the respective descriptors
 - Calculating an homography from the matched keypoints
- Which features are "good features"



Outline

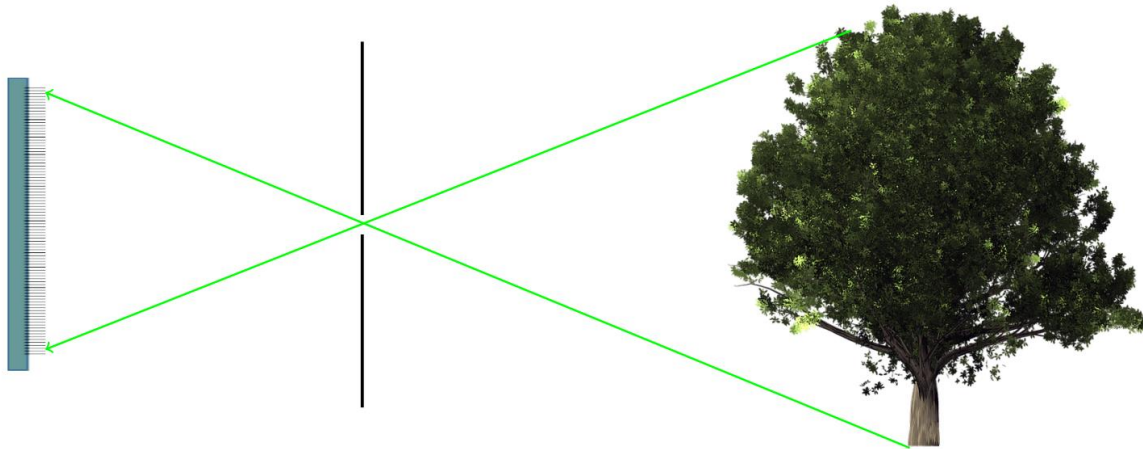
- Cameras and projection
- Camera pose: extrinsic matrix
- Camera calibration: intrinsic matrix
- Direct linear transformation
- Multiplane calibration

Motivation

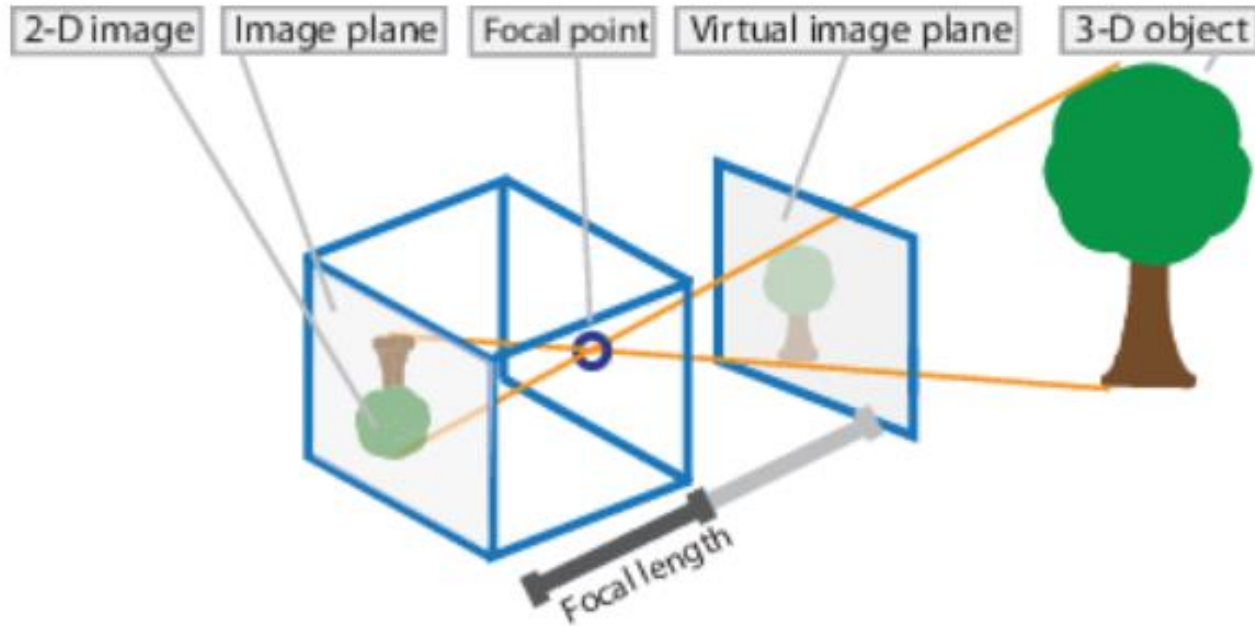
- **Geometric camera calibration** finds the relation between 3D world coordinates and 2D pixels
- Given a **3D** point in the world, which **pixel** will it correspond to?

What is a camera?

- Device that captures light **projected** from a **3D location**
- The **projected** light is captured/recorded by a medium

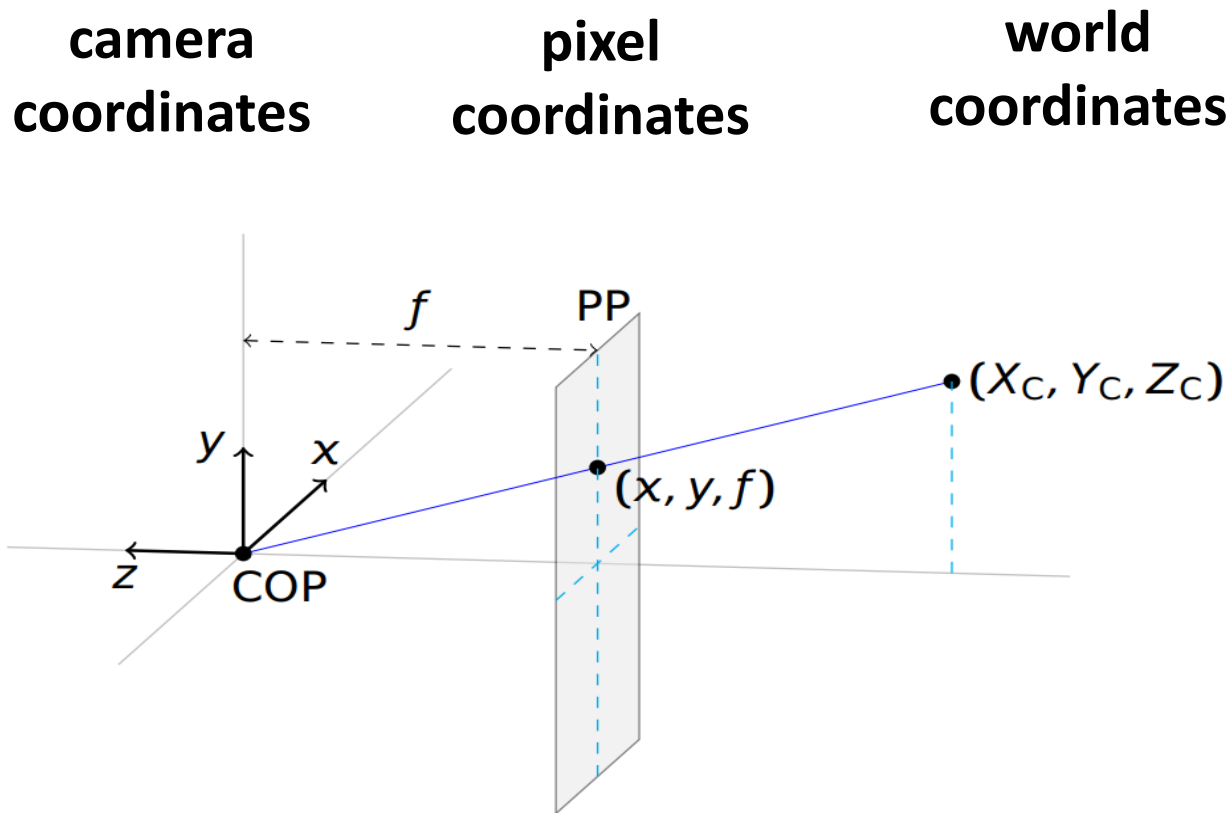


Pinhole Camera Model

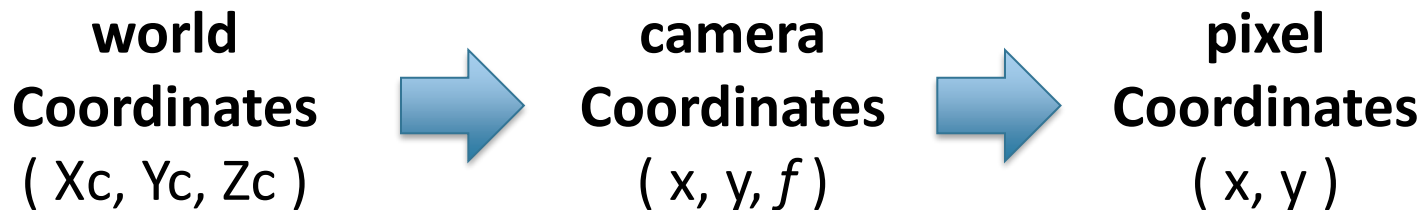


Source: <https://mathworks.com>

Pinhole Camera Model



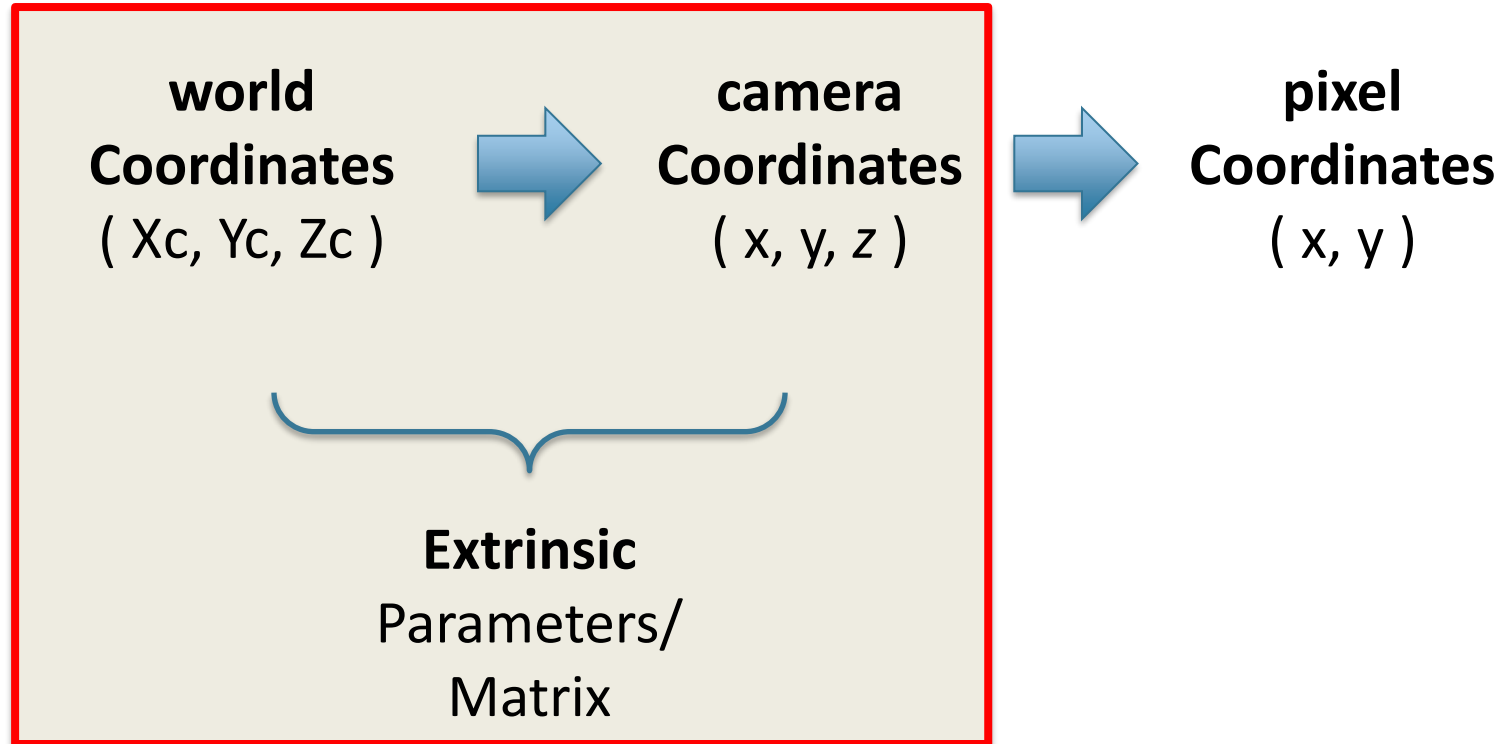
Pinhole Camera Model



Extrinsic
Parameters/
Matrix

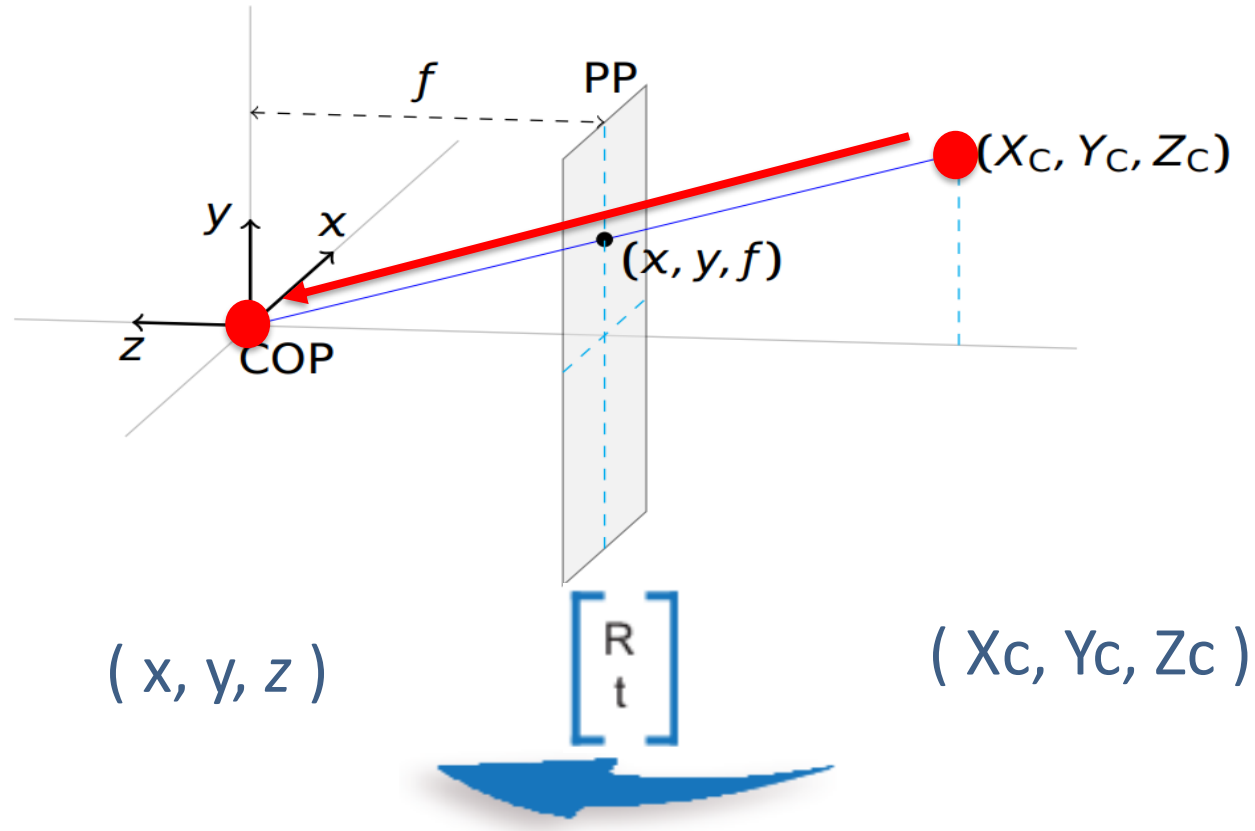
Intrinsic
Parameters/
Matrix

Extrinsic Parameters (M)

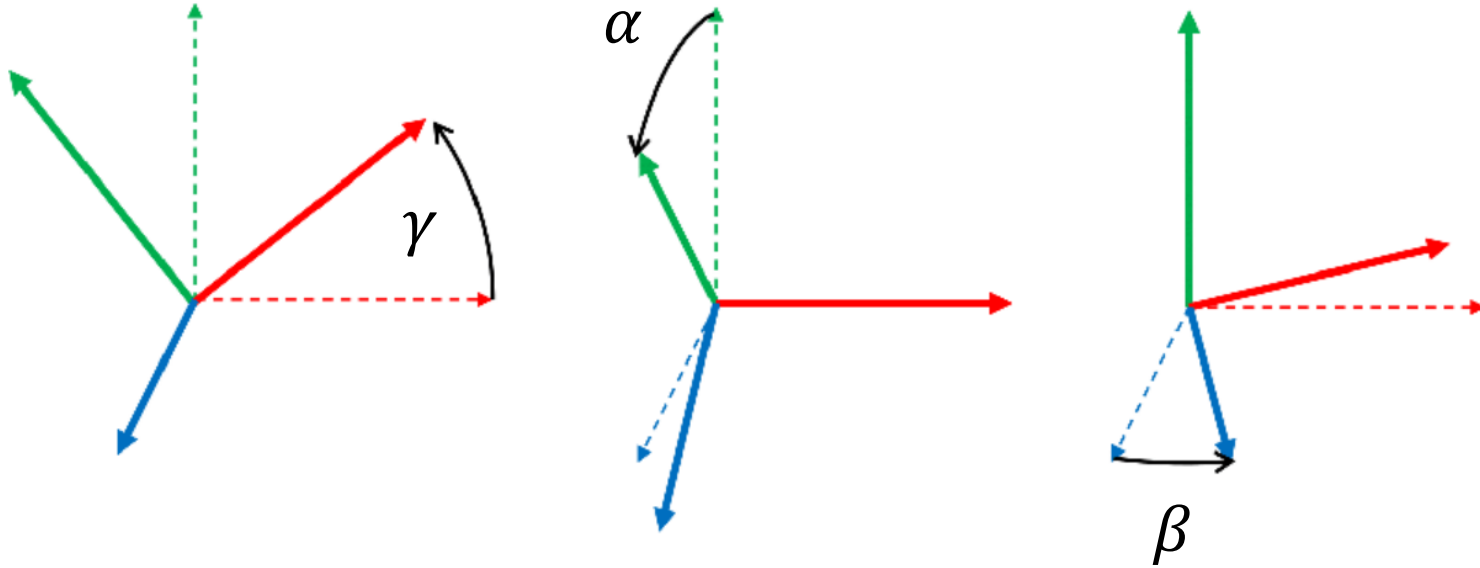


Extrinsic Parameters

- Goal:



Extrinsic Parameters: Rotation



$$R = R_x(\alpha) \cdot R_y(\beta) \cdot R_z(\gamma)$$

Extrinsic Parameters: Rotation

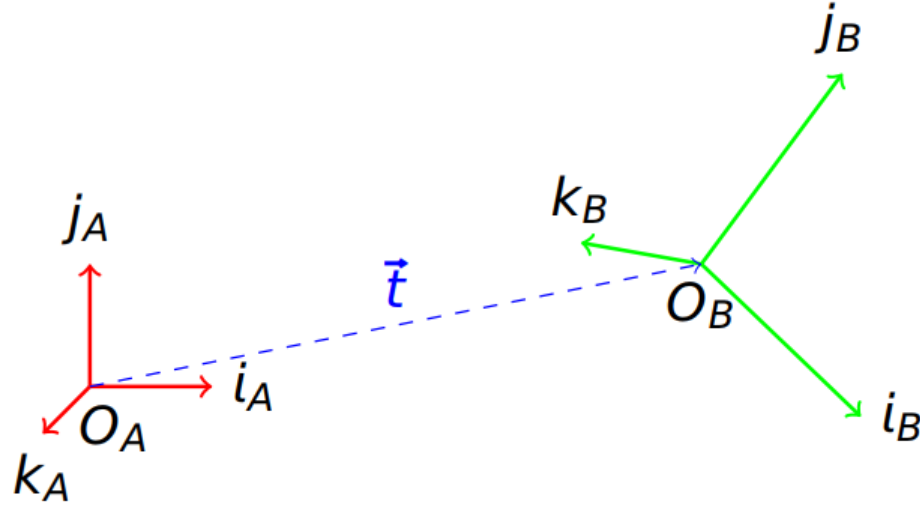
$$R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\alpha) & \sin(\alpha) \\ 0 & -\sin(\alpha) & \cos(\alpha) \end{bmatrix}$$

$$R_y(\beta) = \begin{bmatrix} \cos(\beta) & 0 & -\sin(\beta) \\ 0 & 1 & 0 \\ \sin(\beta) & 0 & \cos(\beta) \end{bmatrix}$$

$$R_z(\gamma) = \begin{bmatrix} \cos(\gamma) & \sin(\gamma) & 0 \\ -\sin(\gamma) & \cos(\gamma) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R = R_x(\alpha) \cdot R_y(\beta) \cdot R_z(\gamma)$$

Extrinsic Parameters: Translation



$$t = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

Extrinsic Parameters: Rotation + Translation

$$t = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}, \quad R = R_x(\alpha)R_y(\beta)R_z(\gamma) = \begin{bmatrix} r_{00} & r_{01} & r_{02} \\ r_{10} & r_{11} & r_{12} \\ r_{20} & r_{21} & r_{22} \end{bmatrix}$$

- Extrinsic matrix (P):
$$P = [R|t] = \begin{bmatrix} r_{00} & r_{01} & r_{02} & t_x \\ r_{10} & r_{11} & r_{12} & t_y \\ r_{20} & r_{21} & r_{22} & t_z \end{bmatrix}$$

Intrinsic Camera Model

**world
Coordinates**
(X_c, Y_c, Z_c)



**camera
Coordinates**
($x, y, z=f$)



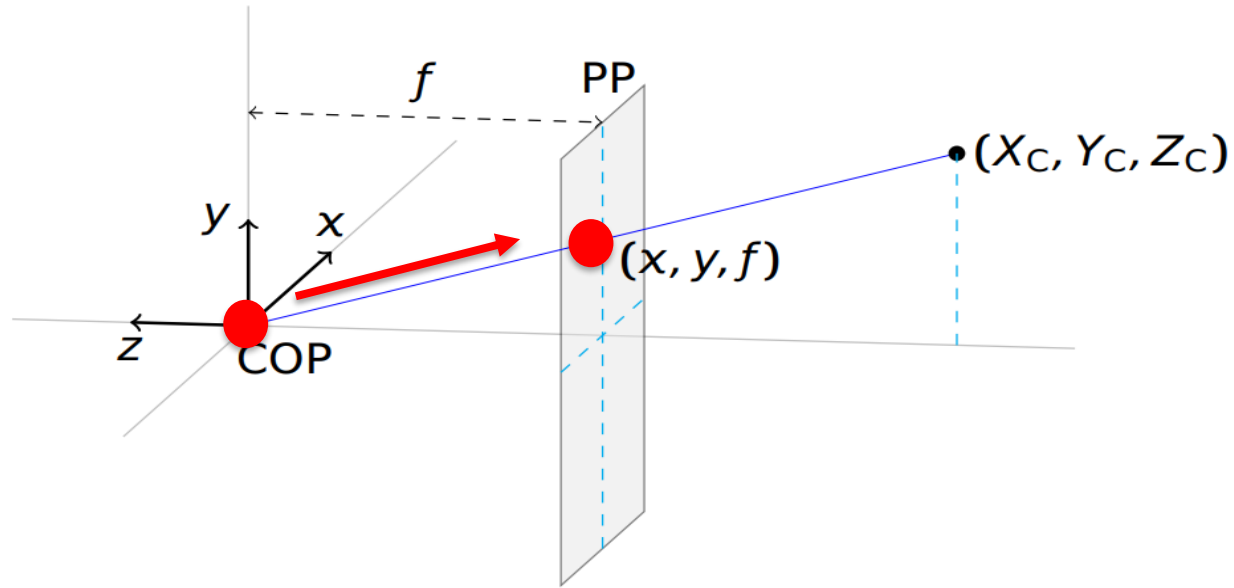
**pixel
Coordinates**
(x_p, y_p)



Intrinsic
Parameters/
Matrix

Intrinsic Parameters

- Goal:

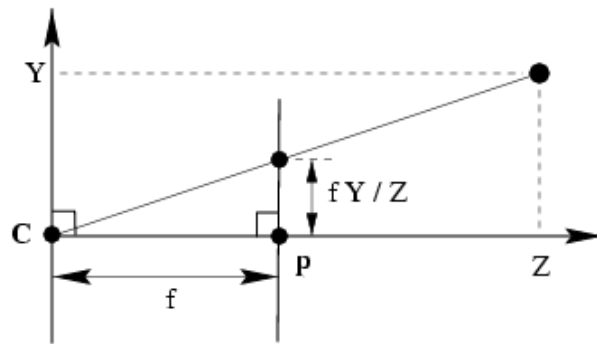
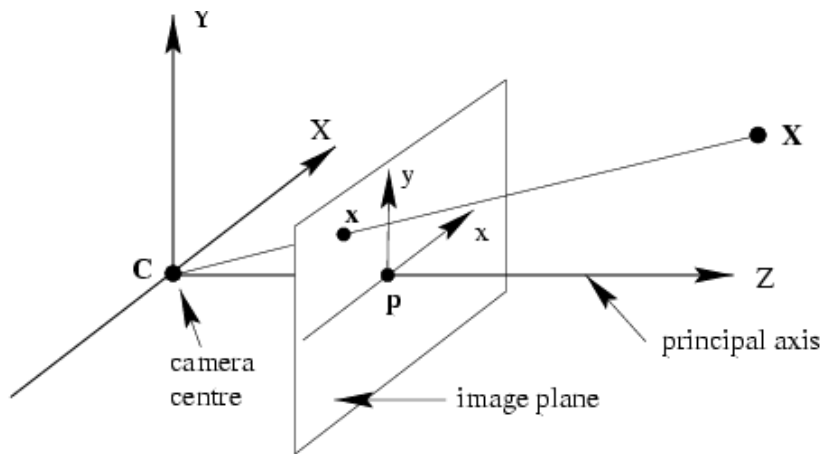


(x, y, z)

(x, y, f)

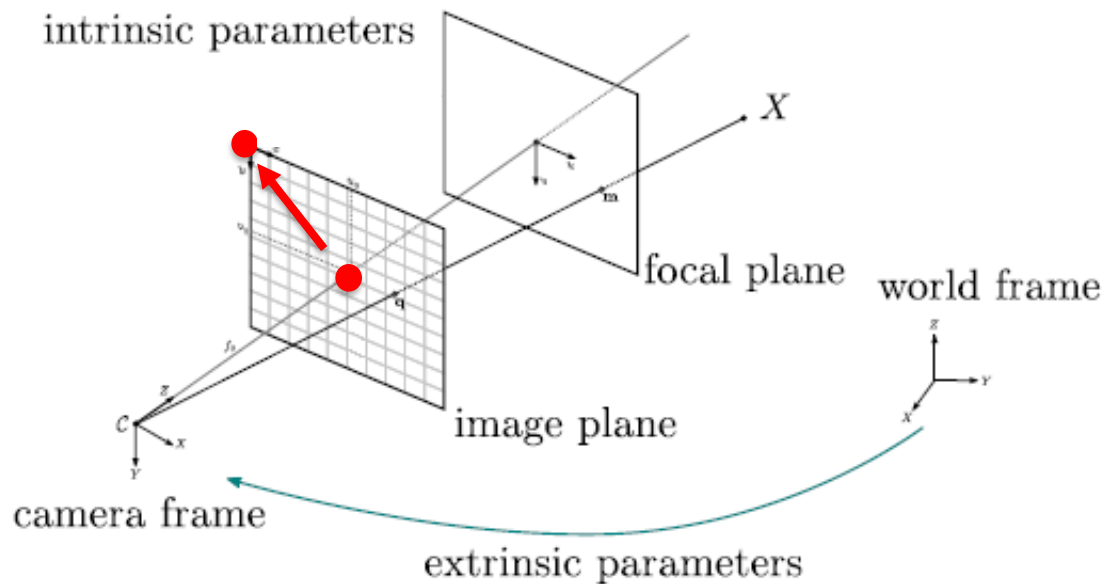


Intrinsic Parameters (K)



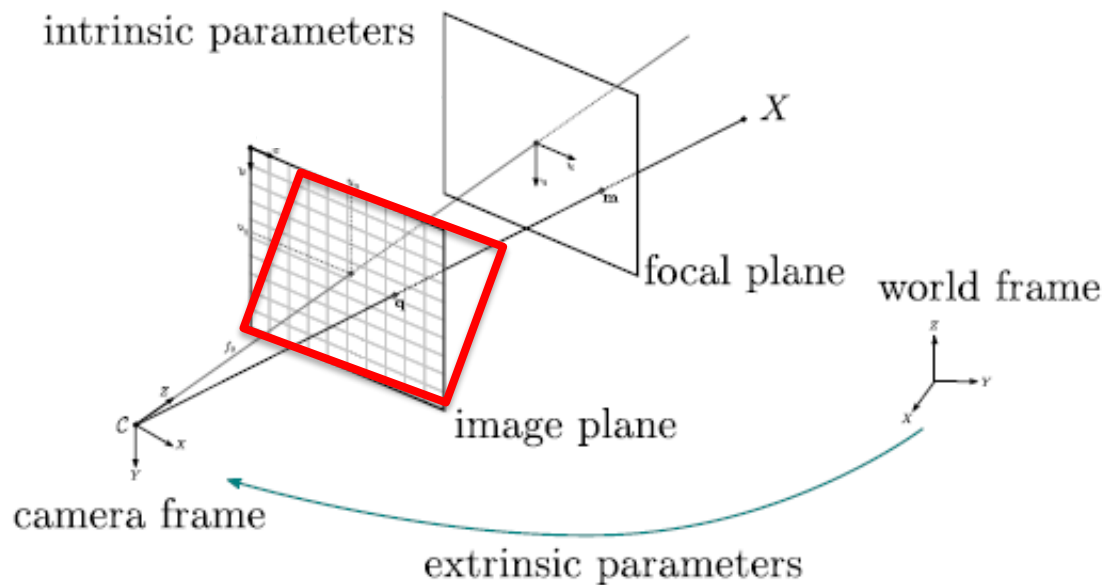
$$K = \begin{bmatrix} f & s & u \\ 0 & af & v \\ 0 & 0 & 1 \end{bmatrix}$$

Intrinsic Parameters: Offset (u,v)



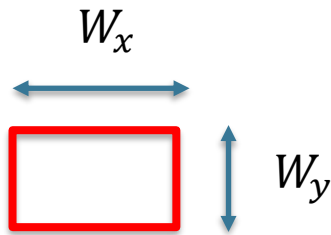
$$K = \begin{bmatrix} f & 0 & \textcolor{red}{u} \\ 0 & f & \textcolor{red}{v} \\ 0 & 0 & 1 \end{bmatrix}$$

Intrinsic Parameters: skew (s)



$$K = \begin{bmatrix} f & \mathbf{s} & u \\ 0 & f & v \\ 0 & 0 & 1 \end{bmatrix}$$

Intrinsic Parameters: aspect ratio (a)



$$a = \frac{W_y}{W_x}$$

$$K = \begin{bmatrix} f & s & u \\ 0 & \textcolor{red}{a}f & v \\ 0 & 0 & 1 \end{bmatrix}$$

Full camera matrix (M)

- Camera matrix M

$$M = K * P = \begin{bmatrix} f & s & u \\ 0 & af & v \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} r_{00} & r_{01} & r_{02} & t_x \\ r_{10} & r_{11} & r_{12} & t_y \\ r_{20} & r_{21} & r_{22} & t_z \end{bmatrix}$$

$$M = \begin{bmatrix} m_{00} & m_{01} & m_{02} & m_{03} \\ m_{10} & m_{11} & m_{12} & m_{13} \\ m_{20} & m_{21} & m_{22} & m_{23} \end{bmatrix}$$

How can we “set” matrix M?

$$M = K * P = \begin{bmatrix} f & s & u \\ 0 & af & v \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} r_{00} & r_{01} & r_{02} & t_x \\ r_{10} & r_{11} & r_{12} & t_y \\ r_{20} & r_{21} & r_{22} & t_z \end{bmatrix} = \begin{bmatrix} m_{00} & m_{01} & m_{02} & m_{03} \\ m_{10} & m_{11} & m_{12} & m_{13} \\ m_{20} & m_{21} & m_{22} & m_{23} \end{bmatrix}$$

M is composed of 2 types of information:

- From scene coordinate system to camera's 3D coordinate system:
Extrinsic parameters (i.e. camera pose); tells me where the camera is in the world
- Find the **intrinsic** parameters (AKA calibration matrix) – from 3D camera coordinate system to 2D image plane via projection

Camera calibration

$$\begin{bmatrix} \acute{x} \\ \acute{y} \\ 1 \end{bmatrix} \approx \begin{bmatrix} w\acute{x} \\ w\acute{y} \\ w \end{bmatrix} = \begin{bmatrix} m_{00} & m_{01} & m_{02} & m_{03} \\ m_{10} & m_{11} & m_{12} & m_{13} \\ m_{20} & m_{21} & m_{22} & m_{23} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$



$$w\acute{x} = m_{00}X + m_{01}Y + m_{02}Z + m_{03}$$

$$w\acute{y} = m_{10}X + m_{11}Y + m_{12}Z + m_{13}$$

$$w = m_{20}X + m_{21}Y + m_{22}Z + m_{23}$$

Camera calibration


$$\begin{aligned} \dot{x} &= \frac{m_{00}X + m_{01}Y + m_{02}Z + m_{03}}{w} = \frac{m_{00}X + m_{01}Y + m_{02}Z + m_{03}}{m_{20}X + m_{21}Y + m_{22}Z + m_{23}} \\ \dot{y} &= \frac{m_{10}X + m_{11}Y + m_{12}Z + m_{13}}{w} = \frac{m_{10}X + m_{11}Y + m_{12}Z + m_{13}}{m_{20}X + m_{21}Y + m_{22}Z + m_{23}} \end{aligned}$$

Camera calibration

- If there are 'n' corresponding 2D-3D points, i.e. $(\acute{x}_i \ \acute{y}_i) - (X_i \ Y_i \ Z_i)$
 $i=1,2,...,n$

$$\begin{bmatrix} X_1 & Y_1 & Z_1 & 1 & 0 & 0 & 0 & 0 & -\acute{x}_1 X_1 & -\acute{x}_1 Y_1 & -\acute{x}_1 Z_1 & -\acute{x}_1 \\ 0 & 0 & 0 & 0 & X_1 & Y_1 & Z_1 & 1 & -\acute{y}_1 X_1 & -\acute{y}_1 Y_1 & -\acute{y}_1 Z_1 & -\acute{y}_1 \\ & & & & & & & \vdots & & & & \\ X_n & Y_n & Z_n & 1 & 0 & 0 & 0 & 0 & -\acute{x}_n X_n & -\acute{x}_n Y_n & -\acute{x}_n Z_n & -\acute{x}_n \\ 0 & 0 & 0 & 0 & X_n & Y_n & Z_n & 1 & -\acute{y}_n X_n & -\acute{y}_n Y_n & -\acute{y}_n Z_n & -\acute{y}_n \end{bmatrix} \begin{bmatrix} m_{00} \\ m_{01} \\ m_{02} \\ m_{03} \\ m_{10} \\ \vdots \\ m_{13} \\ m_{20} \\ \vdots \\ m_{23} \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$AM = 0$$

How can we find corresponding points?

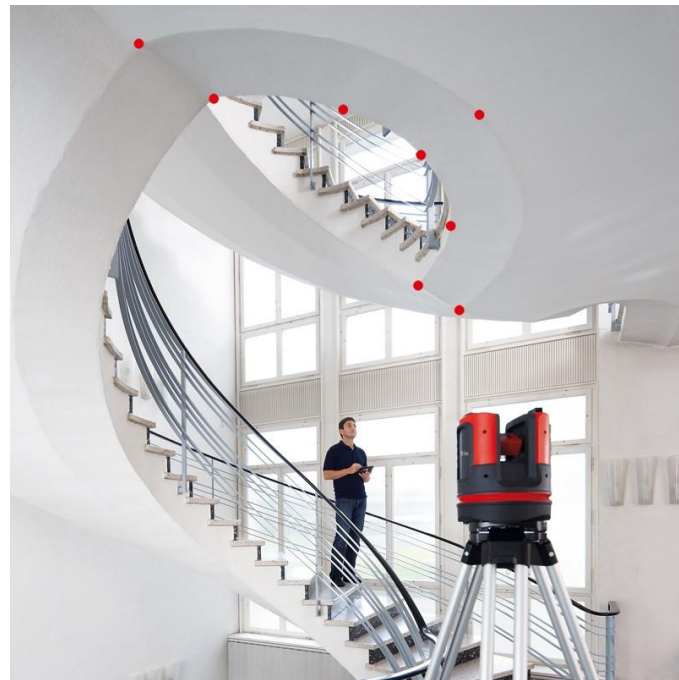
Two popular approaches:

- 1- Direct Linear Calibration

- 2- Multiplane Calibration

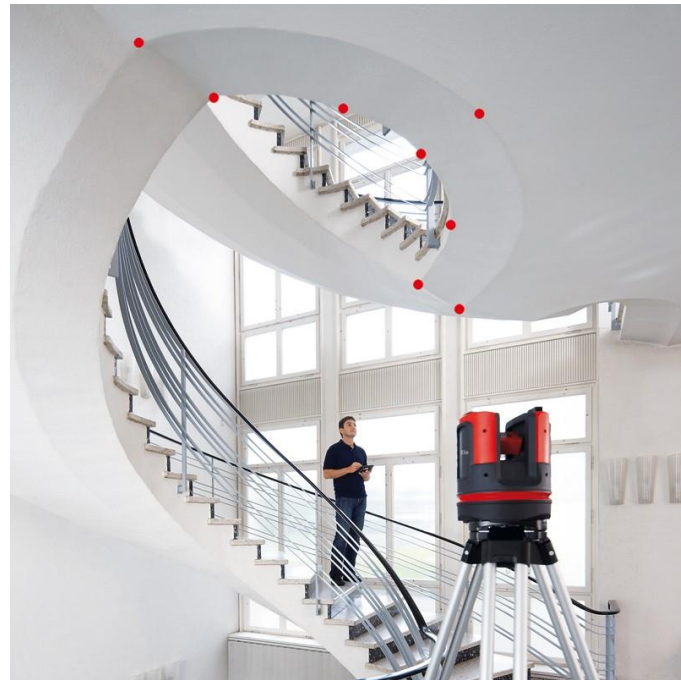
Direct linear calibration

- Needs precise coordinates of points in the world
 - ▶ Provided by measuring locations in scene
 - ▶ Or, by known 3D calibration target

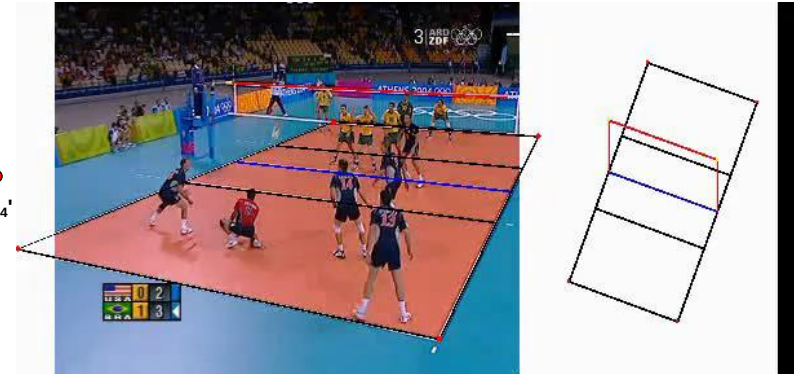
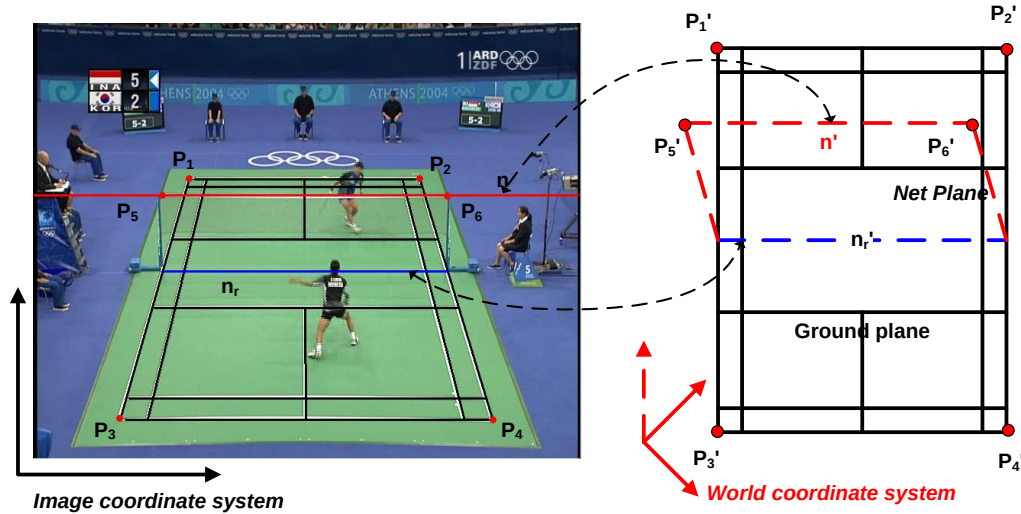


Direct linear calibration

- Needs precise coordinates of points in the world
 - ▶ Provided by measuring locations in scene
 - ▶ Or, by known 3D calibration target
- Problem: must know geometry very accurately



Direct linear calibration Example: Court-Net Calibration



$$\begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} m_{00} & m_{01} & m_{02} & m_{03} \\ m_{10} & m_{11} & m_{12} & m_{13} \\ m_{20} & m_{21} & m_{22} & m_{23} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} \approx \begin{pmatrix} f_x & 0 & o_x \\ 0 & f_y & o_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} r_{00} & r_{01} & r_{02} & t_x \\ r_{10} & r_{11} & r_{12} & t_y \\ r_{20} & r_{21} & r_{22} & t_z \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

Projecting Matrix

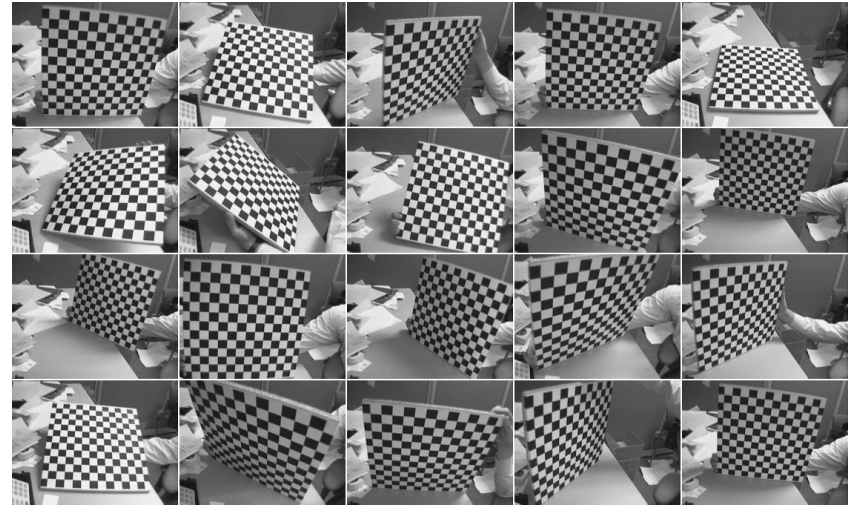
Camera Project

Camera Rotation
and Placement

World
Coordinate

Multiplane calibration

- Proposed by: Zhang (1998)
- Uses a calibration target with a checkerboard pattern
- Pattern must be on a plane with known structure and size
- Uses several pictures of the same target
- Each picture must be in a different location/ orientation/ rotation



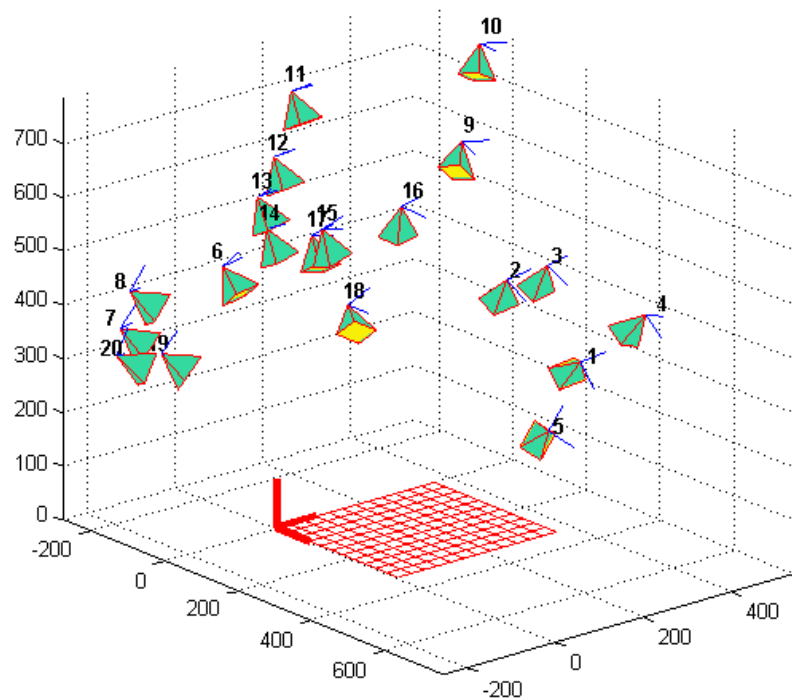
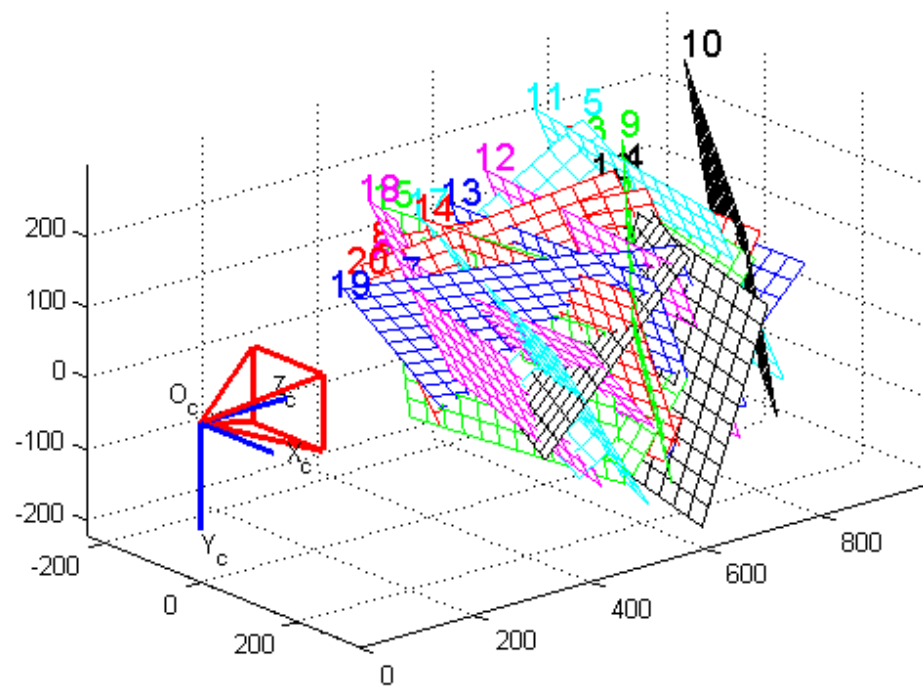
Credit: Wikipedia

Corners

- Corners are detected, we know the $(X, Y, 0)$ of each
- We need to detect, at least 4 points per plane, to produce an homography
- We need a minimum of 3 different views of the plane
- Intrinsics are found by solving a linear system of equations with 6 DoF (each plane gives us 2 equations)



Same intrinsics but different extrinsics per shot



Summary

Direct linear calibration (DLT):

- ▶ Simple to formulate and solve
- ▶ Requires known location in the scene
- ▶ Intrinsics and extrinsics are “mixed” together (requires extra calculation to separate)
- ▶ Won't work if all points are in the same plane in the world

Summary

Multiplane calibration (non-linear):

- ▶ Only requires a simple planar calibration target
- ▶ No need for known world positions
- ▶ Allows imposition of constraints
- ▶ Intrinsics and extrinsics are separated
- ▶ Most popular method plane in the world

References

- Szeliski, 6.2 – 6.5
- Wolfgang Hugemann, Correcting Lens Distortions in Digital Photographs, 2010